



Determination of crop water requirements and potential evapotranspiration for sustainable coffee farming in response to future climate change scenarios

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ARTICLE INFO

Keywords:

Climate change

Coffee

Downscaling

Crop water requirement

Potential evapotranspiration

ABSTRACT

Climate change (CC) is causing a significant threat to agriculture, a sector complicatedly tied to natural resources. Changes in precipitation patterns, atmospheric water content, and rising temperatures intensely affect global agriculture, especially in tropical regions. In this intense CC scenario, potential evapotranspiration (PET) and crop water requirement (CWR) are critical components of agricultural water management. This study evaluates the future impact of CC on precipitation, CWR, and PET in different provinces of Thailand's northern and northeastern regions. Three bias correction methods (Delta (DT), Empirical Quantile Mapping (EQM), and Quantile Mapping (QM)) were employed for precipitation downscaling from the CanESM5 CMIP6-GCM across selected 13 coffee farms with different coffee species. Arabica and Robusta coffee were carefully selected for this analysis. The DT method demonstrated superiority, exhibiting lower RMSE and higher correlation coefficients than EQM and QM. Farm-specific assessments illuminated water demand's complex dynamics during critical growth stages, showcasing variable CWR and PET. During the blooming stage in N-F1, CWR ranged from 16.7 to 33.7 mm/stage, highlighting the variability in water needs. Projected CC impacts on Arabica and Robusta coffee farms in Chiang Rai and Sisaket presented challenges, emphasizing farm-specific strategies to address potential water deficits or surpluses during critical growth phases. Projected 2023, 2028, and 2033 precipitation demonstrated incongruities with CWR and PET. The findings emphasize the crucial role of farm-specific adaptive strategies in mitigating the impacts of changing precipitation patterns on coffee cultivation.

1. Introduction

Climate change (CC) challenges humanity, affecting various sectors such as the agriculture industry and services [1,2]. The central emphasis when evaluating the pivotal dimension of 21st-century CC revolves around examining significant changes in global warming. Various research endeavors have substantiated the increase in the mean surface temperature. [3]. According to the Intergovernmental Panel on CC

(IPCC) report, temperatures on land and ocean surfaces exhibited an average increase ranging from 0.3 to 0.6 °C between 1900 and 1995, with a discernible upward trend of approximately 0.2–0.3 °C over 40 years [4]. Over centuries, the Earth's climate has experienced a gradual warming, marked by a 0.74 °C increase in average temperature [1]. The IPCC (2018) predictions suggest a continued upward trajectory in global warming [5]. To access these predictions, Global climate models (GCMs) are extensively employed across various disciplines, including

List of Abbreviations: CC, Climate Change; CWR, Crop Water Requirement; PET, Potential Evapotranspiration; DT, Delta method; EQM, Empirical Quantile Mapping; QM, Quantile Mapping; GCM, Global Circulation Model; RMSE, Root Mean Square Error; MBE, Mean Bias Error; IPCC, Intergovernmental Panel on Climate Change; CMIP6, Coupled Model Intercomparison Project Phase 6; CMIP5, Coupled Model Intercomparison Project Phase 5; ETC, Crop Evapotranspiration; Eto, Reference Evapotranspiration; Kc, Crop Coefficient; FAO, Food and Agriculture Organization; CDF, Cumulative Distribution Function; TMD, Thai Meteorological Department; ESGF- WCRP, Earth System Grid Federation – World Climate Research Programme.

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<https://doi.org/10.1016/j.atech.2024.100435>

Received 6 February 2024; Received in revised form 8 March 2024; Accepted 8 March 2024

Available online 11 March 2024

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hydrology, biology, and environmental studies, to make future climate projections. In hydrology, GCMs frequently serve as input data for hydrological models, enabling assessments of CC effects on runoff [6], sediment yield [7], precipitation, and temperature [8]. In this development, the 6th phase of the Coupled Model Intercomparison Project (CMIP6) introduces notable advancements compared to its predecessor, CMIP5 [9]. These enhancements are primarily attributed to quantifying radiative forcing, whether natural or human-induced, and incorporating aerosol and land-use effects, as verified by [10,11]. Although CMIP6 data are relatively recent, there is a growing interest in applying the latest projections, as demonstrated by studies such as those by [12–14]. This research observes CC's impacts on agriculture, particularly coffee production in Thailand. The agricultural sector is poised to experience considerable impacts from CC, primarily due to the susceptibility of crops to rising temperatures and water scarcity [15]. Observable harmful outcomes include diminishing crop yields and quality and heightened pest and disease infestations, collectively contributing to a global reduction in crop output [16]. These challenges present substantial hurdles for smallholder farmers, who rely on rain-fed farming and face restricted CC and financial and technical assistance necessary for adapting to shifting climatic conditions [17]. Water is the fundamental factor influencing plant growth, making the precise estimation of water volume essential for various growth stages of crops [18]. Coffee is an essential global commodity, significantly supporting the livelihoods of numerous small-scale farmers worldwide. Being a perennial crop sensitive to climate conditions, coffee is notably vulnerable to the effects of CC [19]. CC also significantly threatens global agriculture, impacting crop yields and quality. Smallholder coffee farmers, particularly in Thailand, face challenges due to rising temperatures and water scarcity, requiring effective adaptation strategies to sustain their livelihoods [20]. Thailand's annual demand for coffee beans hovers around 79,000 tons [21]. Numerous studies have indicated that CC is poised to influence regions suitable for coffee cultivation, particularly impacting Arabica and Robusta coffee and its genetic resources [15–17,22,23]. A comprehensive understanding of crop water requirement (CWR) empowers us to optimize yields by mitigating potential issues associated with excessive or inadequate irrigation practices and worse CC [24]. These issues include waterlogging, soil salinization, and plant water stress, which can significantly impact crop yield outcomes [25]. Through systematic irrigation scheduling, agricultural producers can effectively sustain soil moisture levels above the critical permanent wilting point while optimizing water conservation efforts by eliminating unnecessary irrigation occurrences [25]. The implementation of coffee irrigation scheduling emerges as a promising approach for enhancing yield and facilitating the expansion of coffee plantations, particularly in regions traditionally regarded as unsuitable due to water scarcity constraints [26]. Effective planning and management of irrigation needs a sound understanding of potential evapotranspiration (PET), CWR, and net irrigation requirements. It is essential to recognize that water demands for crops exhibit notable variations among different crop types and throughout the entire growing season of a single crop. This variation underscores the importance of accurate data and information in the decision-making process related to water resource allocation for agricultural purposes [27].

In the Agroecological context of Thailand, understanding the implications of CC on coffee production is crucial for addressing potential threats to a vital global commodity [28,29]. This research seeks to provide insights into the water requirements of coffee cultivation in the face of changing climatic conditions, aiding in the development of sustainable agricultural practices. It is essential to note that while the minimum annual CWR for coffee should fall within the range of 1200–1600 mm, aligning with the study by TesfayeRazi and Maziah, (2008), there exists a need for a precise determination of the irrigation water quantity. This is a must since the annual rainfall in this region often falls short of meeting coffee's seasonal water demands, and the rainfall distribution is typically erratic. Therefore, supplemental

irrigation becomes essential to bridge the water deficit and ensure consistent and reliable support for coffee crop growth and productivity in this specific Agro-ecological setting [30]. The CWR can be accurately computed using a well-established method that estimates crop evapotranspiration (ETc). ETc is determined by multiplying the reference evapotranspiration (ETo) by the crop coefficient (Kc). On a global scale, the Penman-Monteith equation is one of the most precise and comprehensive techniques for calculating evapotranspiration and crop water requirements. This equation, employed in CROPWAT 8.0, is a valuable tool for assessing potential evapotranspiration [31]. The Penman-Monteith equation considers several critical parameters, including temperature, wind speed, humidity, solar radiation, and ambient temperature, to estimate potential evapotranspiration comprehensively. Utilizing the CROPWAT 8.0 model makes it possible to monitor and analyze crop water requirements throughout the growing season, considering both rainfall and supplemental irrigation, thereby facilitating effective water resource management in the field [32]. CWR relates to the volume of water needed to balance for evapotranspiration losses from the cultivated field during a specific time. Typically, these requirements are expressed in mm per day, mm per month, or mm per growing season. Enhancing agricultural practices hinges on adopting contemporary and efficient irrigation strategies and improved water management techniques. When deciding on water utilization, factoring in the prevailing global water situation is imperative. CROPWAT, a user-friendly software model developed by the Food and Agriculture Organization (FAO), is pivotal in determining ETo for precise irrigation scheduling and design. This model relies on the Penman-Monteith Equation formulated by FAO to estimate ETo accurately. The application of CROPWAT not only assists in optimal water management but also contributes to the augmentation of crop production by ensuring that the right amount of water is supplied at the right time for agricultural needs [27].

The primary goal of this research is to assess the impact of CC on coffee cultivation in Thailand and to determine crop water requirements (CWR) and potential evapotranspiration (PET) for coffee cultivation under future CC scenarios. Lastly, projected precipitation (2023–2050) using bias correction methods (DT, EQM, and QM) and compared water availability with CWR and PET at various stages of coffee growth for 2023, 2028, and 2033. The paper is structured as follows: The "Materials and Methods" section outlines the geographical scope, encompassing relevant data on coffee, weather, and soil. It provides a comprehensive overview of the research methodology. The subsequent "Results and Discussion" section presents the outcomes of BC methods, the methodology for determining CWR, and the assessment of PET along with performance evaluations. The concluding section offers final insights, summarizing key findings and their implications.

2. Methodology

2.1. Study area

This research is conducted in the northern and northeastern regions of Thailand. The initial selection of three provinces in Thailand was undertaken, comprising Chiang Rai and Nan in the northern region and Sisaket in the northeastern region. Chiang Rai, located in northern Thailand with an elevation of 390 m, shares borders with Myanmar and Laos. It experiences distinct seasons: summer from mid-March to mid-May, a rainy season until mid-October, and a winter season from mid-October to February. With an average annual rainfall of 1354 mm and maximum and minimum temperatures of 31.5 °C and 20 °C, respectively, Chiang Rai provides ideal conditions for Arabica coffee cultivation [33]. Nan's climatic profile is representative of a tropical monsoon climate, marked by distinct seasonal variations, including winter, summer, and a monsoon-induced rainy season. Typically, the months of July and August receive the highest precipitation. During the winter months, the region experiences cooler and drier conditions owing to the

influence of the northeast monsoon winds, which prevail from November to February. Subsequently, from mid-February to mid-May, the weather tends to be warmer. The area is characterized by an average annual precipitation of approximately 1200 mm, interpreting it as an optimal environment for coffee cultivation. This favorable climate has positioned Nan as the third-largest coffee producer in northern Thailand as of 2022 [2,34]. Sisaket has an average annual rainfall of around 1200 mm, providing an ideal coffee cultivation environment. This favorable climate has positioned Sisaket as a significant contributor to coffee production in northern Thailand, solidifying its role as a critical region in the coffee industry as of 2022. The two principal coffee species, Arabica and Robusta, including their varieties, i.e., Cantimora, were chosen for investigation. In Chiang Rai, four coffee farms (CR-F1, CR-F2, CR-F3, and CR-F4), as depicted in Fig. 1(a), were carefully designated for the study, while six farms in Nan province (N-F1, N-F2, N-F3, N-F4, N-F5, and N-F6) illustrated in Fig. 1(b), were similarly selected. Furthermore, in Sisaket province, three distinct coffee farms (S-F1, S-F2, and S-F3, Fig. 1(c)) were meticulously chosen. The overview of selected

farms is presented in Table 1.

2.2. Coffee farms data for cwr and pet calculation

The determination of CWR and PET within selected 13 coffee farms across various provinces from 2018 to 2022 utilized the CROPWAT program (version 8) developed by the Food and Agriculture Organization (FAO, 2018). The computational procedures employed in this study followed the methodologies outlined by Allen, Pereira et al. (1998) [35]. Key input parameters included daily climate data, comprising Minimum Temperature ($^{\circ}\text{C}$), Maximum Temperature ($^{\circ}\text{C}$), Sunshine Hours (hrs), Wind Speed (km/day), Relative Humidity (%), Latitude, Longitude, and Altitude. The outputs from the Cropwat 8.0 encompassed Radiation ($\text{MJ}/\text{m}^2/\text{day}$) and potential reference evapotranspiration (ET_0) in mm/day. Additionally, coffee crop data incorporated phenological stage duration, depletion coefficient, root depth, crop coefficient, yield response factor, and crop height. The growth stages for planting both species of coffee (Arabica and Robusta) in Thailand are illustrated in

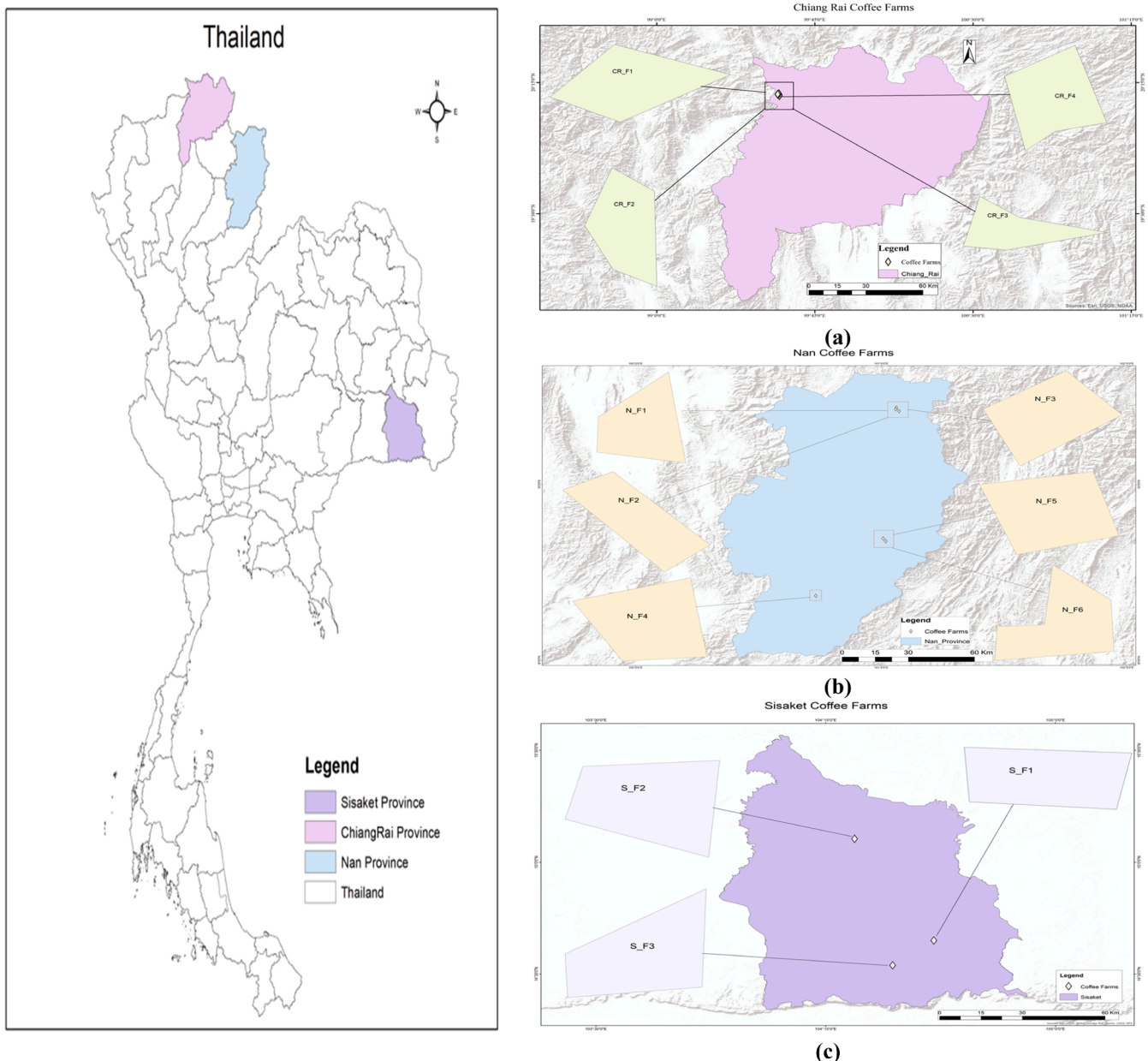


Fig. 1. Selected Coffee Farms in (a) Nan, (b) Chiang Rai, and (c) Sisaket Provinces, Northeastern Region of Thailand.

Table 1

Overview of the Selected Coffee Farm in Northern and Northeastern Thailand.

Coffee Farm	Province	Latitude	Longitude	Sea Level (m)	Species Planted	Plant to Plant distance
N-F1	Nan	19°26'8.34"	101° 3'42.96"	428.8536	Cantimore	1.5 × 1.5 m
N-F2	Nan	19°25'32.75"	101° 3'35.73"	440.436	Cantimore	2 × 2m
N-F3	Nan	19°24'56.21"	101° 4'23.83"	448.3608	Cantimor	1.0–1.5m
N-F4	Nan	18°22'2.00"	100°44'1.90"	387.7056	Robusta (Chumphon 4)	3 × 3 m
N-F5	Nan	18°40'38.03"	101° 1'7.61"	363	Robusta (Chumphon)	2 × 2 m
N-F6	Nan	18°41'33.94"	101° 0'30.80"	316	Robusta (Chumphon)	2 × 2.5 m
CR-F1	Chinag Rai	20° 10.949'	99° 34.900'	407.5176	Arabica	2 × 2 m
CR-F2	Chinag Rai	20° 10.448'	99° 34.948'	347.7768	Arabica	2 × 2 m
CR-F3	Chinag Rai	20° 11.132'	99° 34.754'	377.3424	Arabica	2 × 2 m
CR-F4	Chinag Rai	20° 10.927'	99° 34.427'	457.8096	Arabica	2 × 2 m
S-F1	Sisaket	14°39'1.32"	104°36'11.11"	45.72	Robusta	2 × 3 m
S-F2	Sisaket	14°32'22.81"	104°28'7.45"	56.0832	Robusta	4 × 4 m
S-F3	Sisaket	15° 6'17.49"	104°20'39.93"	40.2336	Robusta	3 × 3 m

Fig. 2. Climate variables such as CWR, PET, and future downscaled precipitation were assessed across all these agricultural stages. Soil data were acquired from the Land and Development Department, Thailand ([https://www.https://www.ldd.go.th/](http://www.https://www.ldd.go.th/)). District-level climate data spanning the years 2018 to 2022 were sourced from the National Aeronautics and Space Administration (NASA) through the Tropical Rainfall Measuring Mission website (<https://pmm.nasa.gov/trmm>) for rainfall and the Prediction of Worldwide Energy Resources website (<http://power.larc.nasa.gov/>) for additional climate variables, including maximum and minimum temperatures, relative humidity, sunshine hours, and wind speed. Agricultural experts can leverage the research methodologies to optimize on-the-ground practices. This study's meticulous statistical downscaling techniques provide a localized perspective, enabling farmers to tailor their cultivation strategies based on projected climate scenarios. Practical applications may include adjusting planting schedules, optimizing irrigation practices, and implementing resilient crop varieties, contributing to enhanced productivity and sustainability.

A comprehensive methodology was designed to achieve the main objectives of this research, as presented in **Fig. 3**. This research aimed to establish a solid foundation by understanding how CC influences coffee cultivation. This process begins with carefully selecting specific coffee farms for in-depth investigation. This research consolidates data from various sources, including the NASA and GCMs derived from the CMIP6 dataset, to capture insights into CC patterns. The amalgamation of these data sources contributes to developing a comprehensive climate dataset, serving as the foundational resource for subsequent analysis. This research employed statistical downscaling techniques to enhance climate data, making it more pertinent at the local level, thus rendering it more suitable for evaluating coffee production. A series of downscaling methods, including Delta, EQM, and QM, are rigorously tested. Subsequently, the most suitable downscaling method is selected based on its performance and applied to all coffee farms within the study. This approach ensures the ability to generate future precipitation projections under various climate scenarios for different timeframes, encompassing 1-year (2023), 5-year(2028), and 10-year(2033) intervals. The selected downscaling method is systematically applied to all coffee farms within the study area, ensuring spatial and temporal relevance for future projection. Notably, the specifics of each technique, including parameters and equations, are thoroughly interpreted to provide a comprehensive understanding of their application. The study addresses uncertainties by meticulously examining underlying assumptions, clearly delineating their potential impacts on results. The CWR and PET for all farms are diligently determined, spanning 2018 to 2022. Additionally, the average CWR and PET at various growth stages of coffee plants are calculated. To provide a comprehensive understanding of the implications of CC, the CWR and PET data about the previously established future precipitation projections are scrutinized. The computation of CWR, indicating the water quantity necessary for crop sustenance, was conducted using the formula $CWR = ET_0 * K_c$, where ET_0 represents Reference Crop

Evapotranspiration and K_c denotes the Crop Coefficient presented in **Eq. (1)** [36]. The ET_0 was determined utilizing the FAO Penman-Monteith equation (Allen et al., 1998). CWR was calculated under standard conditions, assuming disease-free, non-limiting nutrient, and optimal soil water conditions, thereby representing the potential CWR. Given the substantial impact of PET on runoff through surface evaporation, the Penman-Monteith formula used by an investigation [36] was employed for PET calculation in the present study. The overall methodological framework for determining CWR, PET, and analysis is presented in **Fig. 4**. This thorough assessment enables the study to draw informed conclusions regarding the potential impact of CC on coffee production in the selected farms.

$$ET_0 = \frac{0.408\Delta(R_n - G) + \gamma \frac{900}{T+273}(e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)} \quad (1)$$

Where,

ET_0 = Reference evapotranspiration (mm/day)
 R_n = Net radiation at the crop surface (MJ m²/day)
 G = Soil heat flux density (MJ m²/day)
 T = Mean daily air temperature at 2 m height (°C)
 u_2 = Wind speed at 2 m height (m/s)
 e_s = Saturation vapor pressure(kpa)
 e_a = Actual vapor pressure (kpa)
 $e_s - e_a$ = Saturation vapor pressure deficit (kpa)
 Δ = Slope vapor pressure curve (kpa/ °C)
 Γ = Psychrometric constant (kpa/ °C)

The CROPWAT model employs the actual evapotranspiration (ET) and crop coefficient (K_c) for estimating potential evapotranspiration (PET)[36], utilizing the following formula:

$$PET = \frac{ET}{K_c} \quad (2)$$

2.3. Bias correction methods for downscaling

The statistical downscaling of precipitation data for both current and future scenarios was conducted using bias correction methods, namely, DT [37], QM [38], and EQM [39]. A comprehensive understanding of these techniques has been detailed in the literature [37–42].

2.3.1. Delta-Change method

The DT method [43] generates climate scenarios by incorporating the design research signal from GCM predictions as a modification to observational datasets rather than relying directly on GCM simulations of the future.

$$P^{Mod}(t) = P^{obs}(t) \quad (3)$$

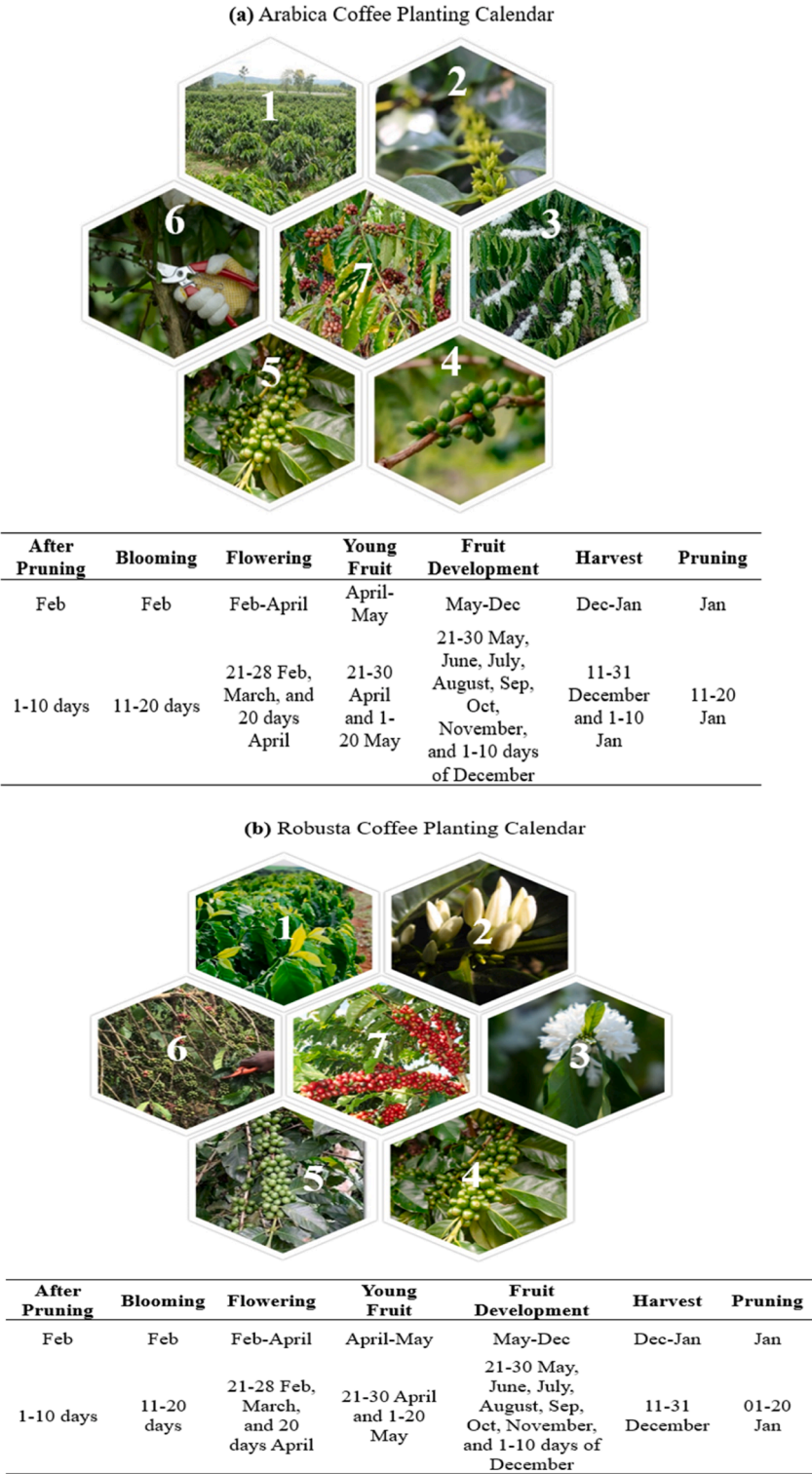


Fig. 2. (a) Arabica Coffee Planting (b) Robusta Coffee Planting (Different growth stages) Calander.

$$P^{frc}(t) = P^{obs}(t) \frac{[\mu d * P^{frc}(t)]}{[\mu d.P^{mod}(t)]}$$
 (4)

"P" represents the time series derived from the forecast requiring adjustment, where "t" denotes the time step, and "d" signifies the long-term daily mean. The only altered variable is "t." The GCM simulations conducted during the control period yield the time series labeled PMod, while the observed data results in the time series referred to as Pobs. The

observed time series are GCM simulations illustrating variations between the controlled environment and the scenario. The multiplicative correction process is typically executed monthly.

2.3.2. Empirical quantile mapping

EQM employs a transfer function (TF) to align the distribution of simulated data from the GCM with that of the observational dataset [44]. The TF serves as the mechanism for this adjustment, comparing

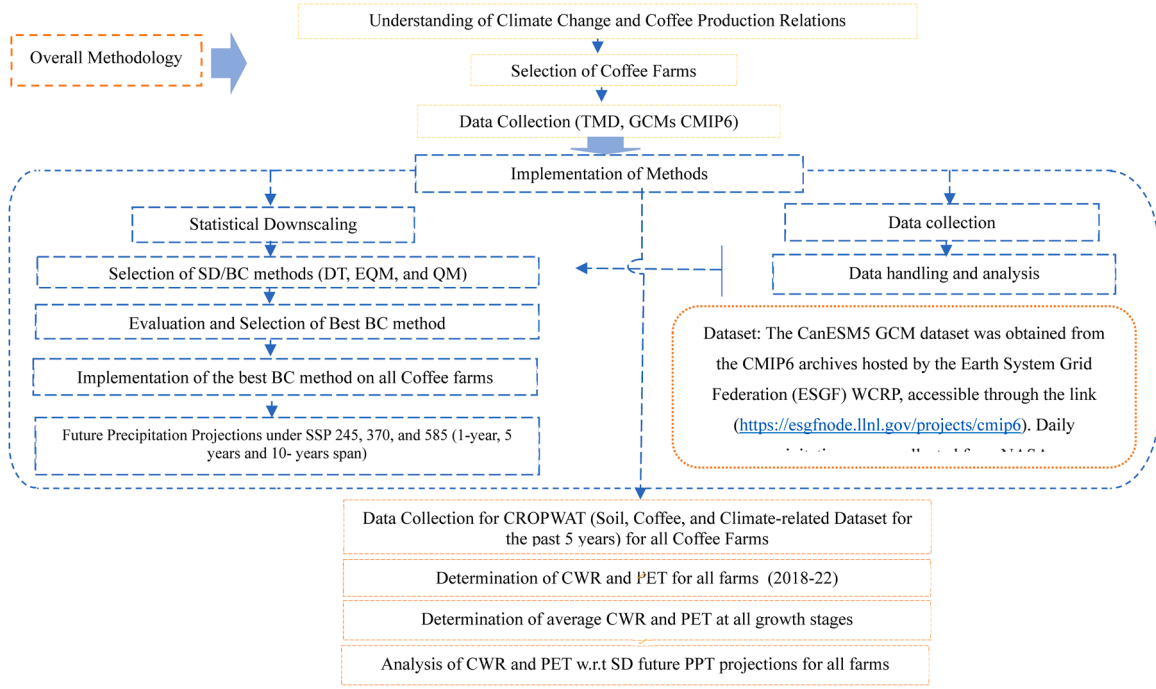


Fig. 3. Overall methodology of the study.

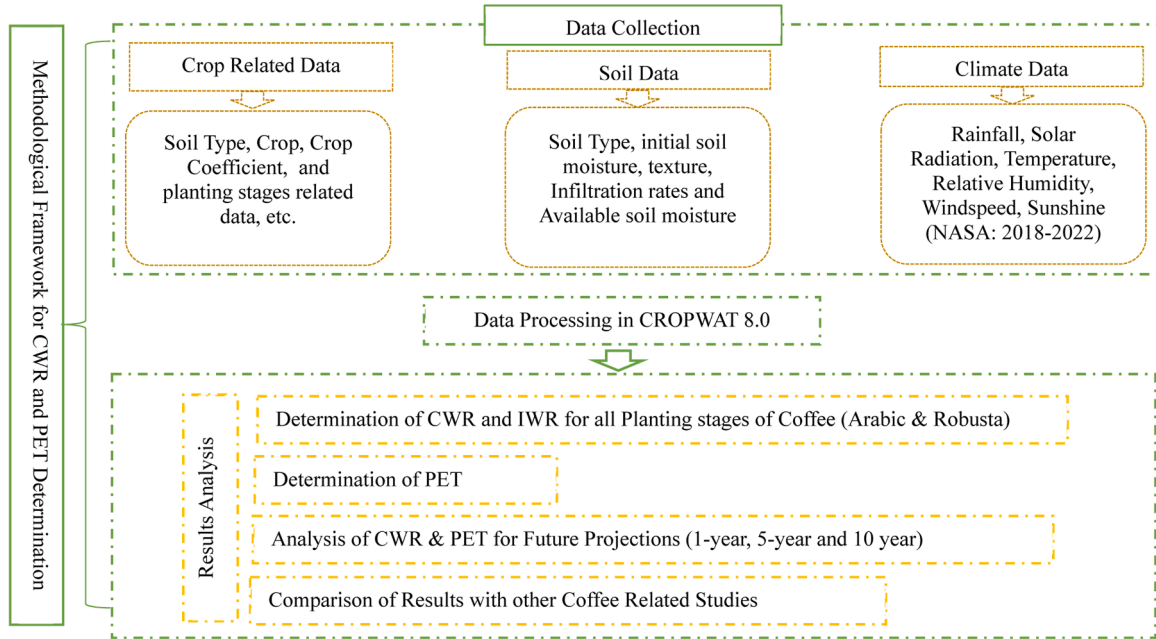


Fig. 4. Methodological Framework for CWR and PET Determination.

observed data with GCM-generated data through statistical moments of each function's cumulative distribution function (CDF).

$$P^{Mod}(t) = CDF^{-1}_{obs, m}(CDF_{mod, m}(P^{obs}(t))) \quad (5)$$

$$P^{frc}(t) = CDF^{-1}_{obs, m}(CDF_{frc, m}(P^{frc}(t))) \quad (6)$$

QM introduces a non-linear correction that preserves extremes while adjusting the data's mean, variance, quantiles, and intensities. This correction aligns the variability of the corrected data more closely with that of the original GCM data.

2.3.3. Quantile mapping

QM was introduced by Panofsky and Brier (1968); QM is widely utilized in statistical downscaling across diverse fields [45]. All GCM-based bias correction methods involve comparing the distribution of simulated climatic variables with locally observed historical data. In applying standard QM, the regularity assumption of biases and redistributive features implies that these variables remain consistent across past and future periods, with only the means potentially changing [46]. QM aligns cumulative distribution functions (CDFs) of GCM and observed data, adjusting GCM projections to better match the statistical characteristics of observed data and enhance downscaled climate

information accuracy. The QM equation involves modeling probabilistic distribution against observed distribution, with specific computations outlined for Standard Deviation (SD) GCM evaluation (Eq. (6)) and future downscaling (Eq. (7)), especially in the context of precipitation data.

$$P_{eva} = \text{InvCDF}_{P_{i-Cul}}^{\text{stat}}(\text{CDF}_{P_{i-Cul}}^{\text{hist}}(P_{i-Eval}^{\text{GCM}})) \quad (7)$$

$$P_{pre} = \text{InvCDF}_{P_{i-Hist}}^{\text{stat}}(\text{CDF}_{P_{i-Hist}}^{\text{hist}}(P_{i-SSP}^{\text{GCM}})) \quad (8)$$

The CDF portrays the cumulative probability distribution of observational and GCM data across a defined period.

2.4. Performance assessment

In the performance evaluation phase, measures such as root mean square error (RMSE), mean absolute error (MAE), Pearson's correlation (r), index of agreement (d), and Mean Bias Error (MBE) are employed to assess the bias between the output of each bias correction method and the observed data.

$$\text{RMSE} = \frac{\sqrt{\sum_{i=1}^N (\text{Pobs} - \text{Ppre})^2}}{N} \quad (9)$$

$$\text{Pearson's correlation (r)} = \frac{n(\sum P_{obs} * P_{pre}) - (\sum P_{obs})(\sum P_{pre})}{\sqrt{[n\sum P_{obs}^2 - (\sum P_{obs})^2][n\sum P_{pre}^2 - (\sum P_{pre})^2]}} \quad (10)$$

$$\text{MBE} = \frac{1}{n} \sum_{i=1}^n (\text{Ppre} - \text{Pobs}) \quad (11)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\text{Pobs} - \text{Ppre}| \quad (12)$$

$$\text{Index of Agreement (d)} = 1 - \frac{\sum_{i=1}^N (P_{obs} - P_{pre})}{\sum_{i=1}^N (|P_{pre} - \bar{P}_{obs}| + |P_{obs} - \bar{P}_{obs}|)} \quad (13)$$

P_{obs} represent observed precipitation, P_{pre} signifies projected precipitation, and P_{mean} denotes the mean of observed precipitation. Pearson's correlation coefficient (r) ranges from -1.00 to $+1.00$, where

-1.00 indicates a perfect negative correlation, $+1.00$ denotes a perfect positive correlation, and 0.00 implies no relationship between the variables [47]. The index of agreement (d) ranges from 0 to 1, with higher values indicating more substantial agreement between predicted values (Ppre) and observed values (Pobs) [48]. A nominal RMSE value suggests a well-performing model [49]. MBE assesses the average bias of the model and determines the need for bias correction. It quantifies the typical prediction bias. MAE represents the difference between expected and observed values [40]. MAE and RMSE indicate average prediction errors in units of the dependent variable, with values from 0 to ∞ and independent of the error direction [50].

3. Results and discussion

3.1. Projection of precipitation for all coffee farms

The results of evaluating three bias correction methods—Delta, EQM, and QM—for statistical downscaling precipitation from the Can-ESM5 CMIP6-GCM across 13 coffee farms in northern and northeastern Thailand reveal compelling insights shown in Fig. 5. Firstly, the RMSE, which is a crucial indicator of overall model performance, the Delta method consistently outshines EQM and QM with substantially lower RMSE values. For instance, in the case of the coffee farm CR-F1, Delta achieves an RMSE of 4.782, while EQM and QM register higher values of 7.749 and 8.194, respectively. This pattern persists across all stations, positioning Delta as the method with the closest fit to the observed precipitation. NRMSE further substantiates the superiority of the Delta method. With NRMSE values consistently lower than those of EQM and QM, Delta demonstrates superior predictive accuracy. For instance, in station N-F1, Delta's NRMSE of 1.015 compares favorably to EQM's 2.223 and QM's 2.244. Examining correlation coefficients, both Pearson and Spearman, Delta maintains commendable values across all stations. In CR-F1, Delta yields a Pearson correlation of 0.601 and Spearman correlation of 0.737, outperforming EQM and QM. This consistent pattern indicates Delta's proficiency in capturing linear and monotonic relationships. MAE analysis underscores Delta's precision in predicting precipitation. Across various stations, Delta consistently records the lowest MAE values, exemplified by CR-F2, where Delta's MAE is 1.52 compared to EQM's 1.61 and QM's 1.59. The Bias (MBE) results reveal Delta's remarkable ability to minimize systematic overestimation or underestimation. For CR-F3, Delta's MBE of 0.0 contrasts sharply with

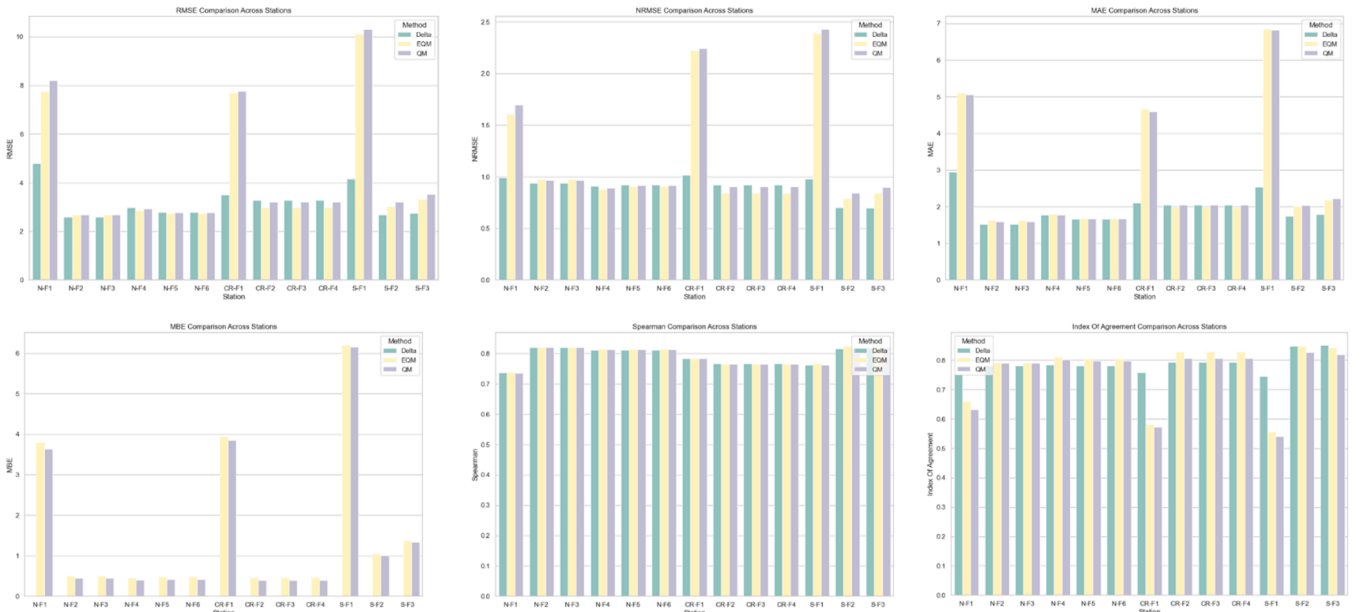


Fig. 5. Evaluation results for bias correction methods (Delta, EQM, and QM) for statistical downscaling at all coffee farms.

EQM's 0.5 and QM's 0.45, showcasing Delta's unbiased performance. The index of agreement (d) gauging agreement between modeled and observed precipitation quantiles further positions Delta as the optimal choice. With consistently high 'd' values, Delta accurately represents quantile information. This comprehensive analysis of statistical metrics unequivocally favors the Delta method for precipitation downscaling. Its superior performance in terms of RMSE, NRMSE, correlation coefficients (Pearson and Spearman), MAE, and bias underscores Delta as the most reliable and accurate method for predicting precipitation across the diverse coffee farms in Thailand.

The downscaling results for monthly average precipitation at various

coffee farms from 2023 to 2050, shown in Fig. 6, provide insightful information about the mean and SD of precipitation patterns. The analysis conducted to assess the impact of CC on precipitation patterns and, consequently, on coffee water requirements reveals intriguing trends across various farms. These trends are particularly evident when examining each farm's mean precipitation and SD values from January to December (2023–2050). Observations from the N-F1 farm indicate a fluctuation in mean precipitation, ranging from 0.24 mm in January to 11.47 mm in June. Similarly, the SD varies from 0.61 mm in January to 16.92 mm in June. This variability underscores the challenges posed by changing precipitation patterns, necessitating adaptive strategies in

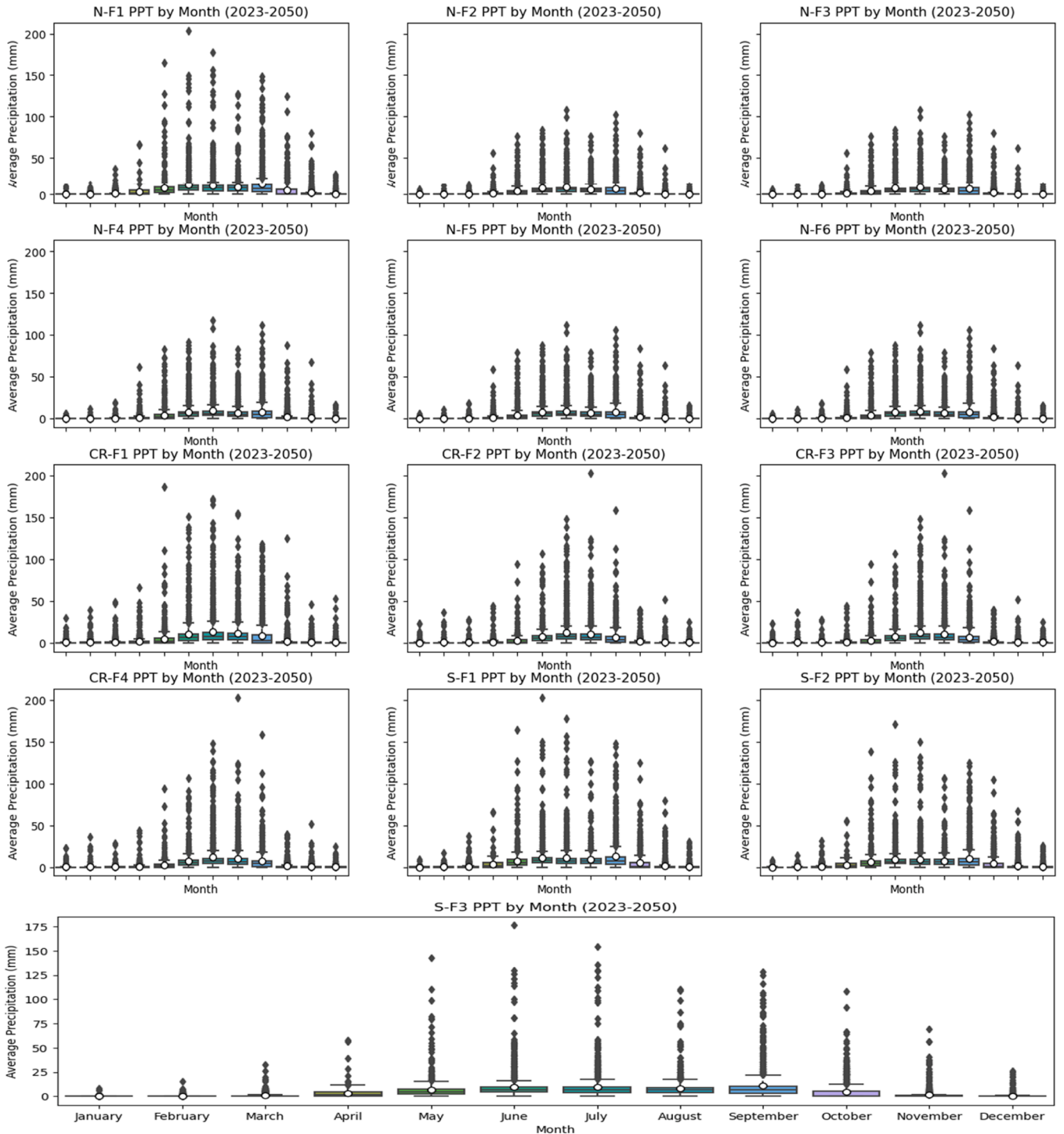


Fig. 6. Monthly precipitation analysis of projections for all coffee farms downscaling by Delta method for years (2023–2050).

water management for coffee cultivation. The CR-F1 farm exhibits a broader range of mean precipitation, from 0.64 mm in January to 13.83 mm in July. The corresponding standard deviation values fluctuate between 1.99 mm and 21.42 mm during the same period. The higher mean precipitation suggests potential water abundance, but the substantial standard deviations emphasize the unpredictability associated with CC. Contrastingly, the N-F4 and S-F1 farms show lower mean precipitation values, indicating potential water scarcity. For N-F4, mean precipitation ranges from 0.14 mm in January to 8.31 mm in June, with standard deviation values fluctuating between 0.48 mm and 11.23 mm. Similarly, S-F1 ranges from 0.24 mm to 11.47 mm for mean precipitation, with standard deviation values ranging from 0.61 mm to 16.92 mm. These farms may face challenges sustaining coffee cultivation, emphasizing the need for adaptive water management practices. The analysis underscores that farms with higher mean precipitation should focus on flexible irrigation schedules to accommodate fluctuations, while those with lower mean precipitation need to implement robust water conservation practices and efficient irrigation systems. Sustainable agriculture practices tailored to specific precipitation patterns can enhance resilience in the face of changing climatic conditions, ensuring the continued viability of coffee cultivation (Fig. 7).

3.2. Analysis of CWR and PET for all coffee farms

The results of the five-year average (2018–2022) of CWR and PET for various growth stages at all selected coffee farms provide valuable

insights into the water dynamics of coffee cultivation. The cultivation of coffee in Nan province predominantly encompasses two principal species, namely Robusta and Cantimora (a hybrid of Arabica and Robusta). For N-F1, as shown in Fig. 7(a), CWR during blooming ranges from 16.7 to 33.7 mm/stage, averaging 26.34 mm/stage, with a 7.92 mm/stage SD. The flowering stages demonstrate a wider range, from 40 to 164.9 mm/stage, averaging 112.5 mm/stage and a substantial SD of 60.86 mm/stage. Fruit development exhibits the highest CWR, ranging from 120.4 to 741.7 mm/stage, averaging 287.32 mm/stage, with a significant SD of 256.41 mm/stage. Harvest and pruning stages show lower CWR values but still reflect variability. Concurrently, PET values mirror these trends with corresponding averages and SDs. In N-F2, as shown in Fig. 7(b), flowering stages stand out with significant CWR fluctuations, ranging from 24.4 to 212.8 mm/stage, averaging 131.2 mm/stage, with a broad SD of 70.29 mm/stage. Fruit development requires the highest CWR, ranging from 138.9 to 915.9 mm/stage, averaging 242.1 mm/stage, and showing a considerable SD of 101.2 mm/stage. N-F3 exhibits diverse CWR and PET trends across growth stages (Fig. 7(c)). During flowering, CWR varies significantly from 36.6 to 206.8 mm/stage, averaging 128.9 mm/stage, with a substantial SD of 63.48 mm/stage. Fruit development requires the highest CWR, ranging from 98.8 to 792.6 mm/stage, averaging 188.06 mm/stage, with a noteworthy SD of 84.17 mm/stage. Harvest stages demonstrate a CWR range of 58.1 to 78.5 mm/stage, averaging 67.58 mm/stage, with a moderate SD of 7.01 mm/stage, for N-F4 as presented in Fig. 7(d), flowering stages exhibit considerable CWR fluctuations, ranging from 96.4 to 270.2 mm/stage,

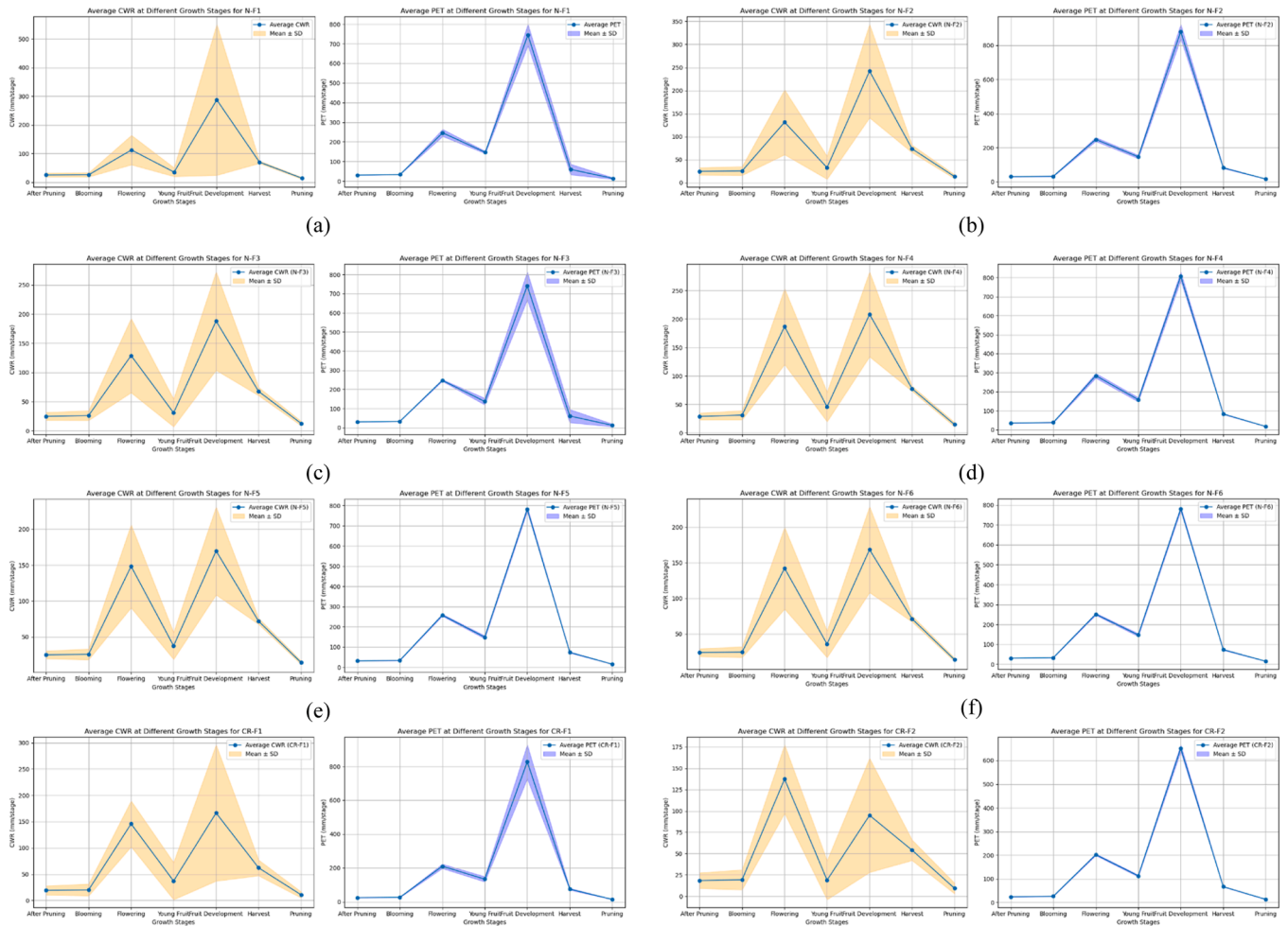


Fig. 7. Analysis of average crop water requirement (CWR) and potential evapotranspiration (PET) across all coffee farms in Nan, Chiang Rai, and Sisaket provinces. The depicted figures (a-m) represent detailed insights into the agricultural dynamics of N-F1 to N-F6, CR-F1 to CR-F4, and S-F1 to S-F3, providing a comprehensive overview of water requirements and evapotranspiration patterns in the specified region.

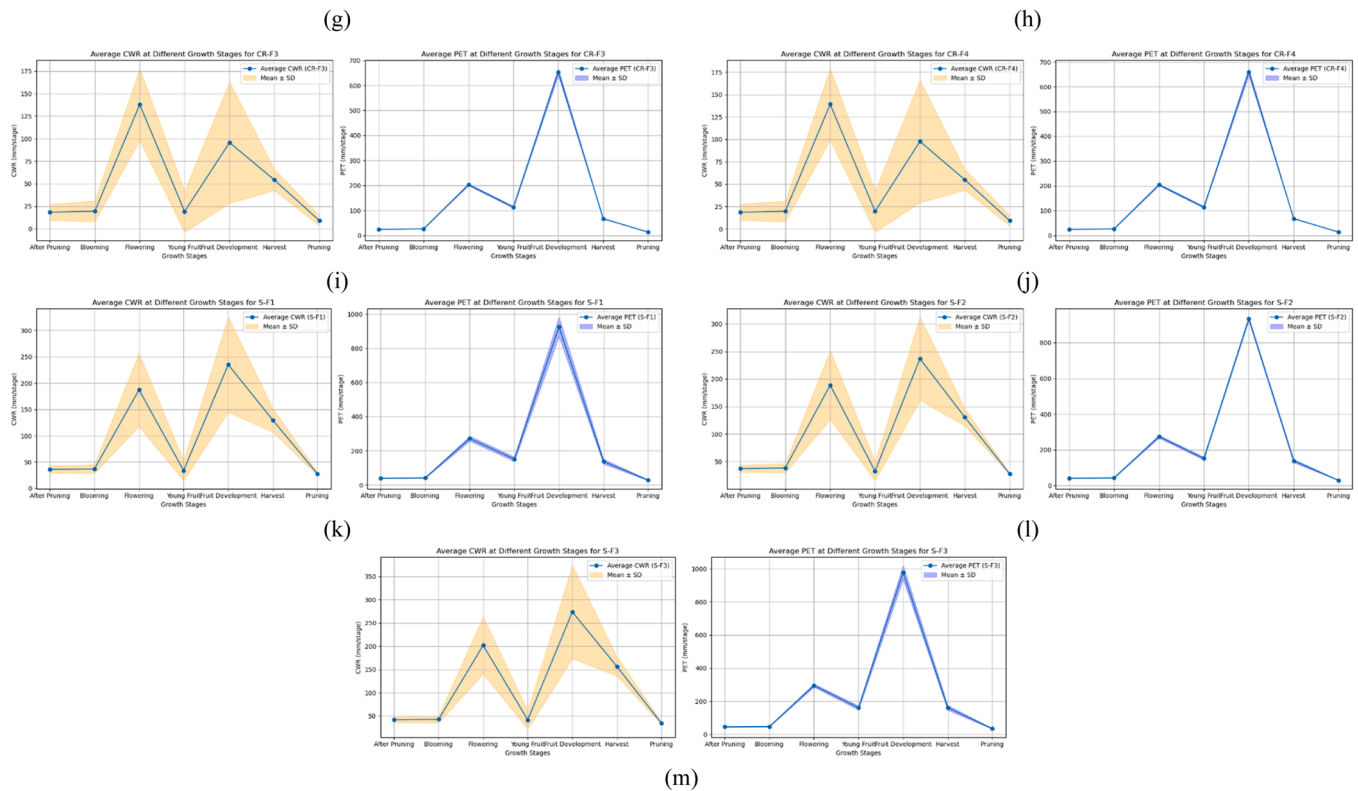


Fig. 7. (continued).

averaging 186.62 mm/stage, with a notable SD of 66.46 mm/stage. Fruit development entails the highest CWR, from 152.8 to 844.2 mm/stage, averaging 208.12 mm/stage, with a distinct SD of 74.41 mm/stage. Harvest stages demonstrate a CWR range of 73.3 to 85.2 mm/stage, averaging 77.6 mm/stage, with a modest SD of 4.27 mm/stage. In N-F5 (Fig. 7(e)), significant variations in CWR are observed during flowering stages, ranging from 59 to 212.8 mm/stage, averaging 148.34 mm/stage, with a considerable SD of 57.82 mm/stage. Fruit development necessitates a CWR range from 124.9 to 798.4 mm/stage, averaging 169.72 mm/stage, with a notable SD of 61.2 mm/stage. Harvest stages demonstrate a CWR range of 67.6 to 78.6 mm/stage, averaging 71.72 mm/stage, with a moderate SD of 4.15 mm/stage. For N-F6, the flowering stages exhibit dynamic CWR fluctuations, ranging from 54.1 to 206.6 mm/stage, averaging 142.4 mm/stage, with a distinct SD of 57.29 mm/stage Fig. 7(f). Fruit development, marked by a CWR range from 125.2 to 795.1 mm/stage, manifests an average of 168.64 mm/stage and a notable SD of 60.27 mm/stage. Harvest stages showcase a CWR range of 67 to 78.1 mm/stage, averaging 71.36 mm/stage, with a modest SD of 4.2 mm/stage. The dynamic interaction between CWR and PET across various growth stages underscores the need for adaptive water management strategies. These insights are crucial for sustaining optimal crop health and productivity in coffee farms, ensuring they thrive in diverse environmental conditions.

By using the same methodology at coffee farms in Chiang Rai (CR-F1, CR-F2, CR-F3, and CR-F4), the CWR and PET dynamics across growth stages at all coffee farms, discernible patterns emerge. In Fig. 7(g) for CR-F1, the flowering stage exhibits a substantial CWR range from 100.3 to 211.9 mm/stage, averaging 145.7 mm/stage with a notable SD of 43.57 mm/stage. Fruit development stages demonstrate a wide-ranging CWR, varying from 51.4 to 900.3 mm/stage, indicating significant variability in water demand during this phase. Harvest stages, characterized by a CWR range of 48.7 to 80 mm/stage, underscore the importance of precision irrigation strategies. Moreover, the blooming stage presents an interesting juxtaposition, with a CWR range of 1.9 to

28.1 mm/stage, highlighting the nuanced water requirements during this critical phase. Examining the CR-F2 coffee farm's growth stages reveals noteworthy insights into CWR and PET, as shown in Fig. 7(h). The flowering stage stands out with a substantial CWR range of 89.7 to 198.6 mm/stage, averaging 137.32 mm/stage and exhibiting a relatively low SD of 40 mm/stage. Contrastingly, the blooming stage showcases a diverse CWR range from 0.6 to 27 mm/stage, emphasizing the dynamic water needs during this phase. Fruit development stages demonstrate considerable variability, with CWR ranging from 33.1 to 669.2 mm/stage, underscoring the importance of targeted irrigation practices. The harvest stage, with a CWR range of 39.3 to 68.6 mm/stage, highlights the need for precision water management during this critical period. The flowering stage exhibits a PET range from 198.4 to 206.8 mm/stage, averaging 202.4 mm/stage, with a relatively low SD of 3.69 mm/stage. This indicates a consistent and moderate evapotranspiration demand during the flowering phase. Fruit development stages reveal a broader PET range, from 634.3 to 669.2 mm/stage, emphasizing the substantial atmospheric demand for water in supporting fruit maturation. The harvest stage, with a PET range of 65.3 to 81.84 mm/stage, signifies a moderate atmospheric water demand during this crucial period. Similarly, the contrasting patterns of CWR and PET in CR-F3 and CR-F4 coffee farms underscore the nuanced water dynamics in these agricultural settings, as presented in Fig. 7(i) and 7(k). In CR-F3, the flowering stage demonstrates a substantial CWR range of 90.3 to 137.88 mm/stage, reflecting a dynamic water demand during this critical phase of plant development. The corresponding PET, ranging from 199 to 207.4 mm/stage, indicates a consistent atmospheric demand, emphasizing the importance of precise irrigation management during flowering. On the other hand, CR-F4 exhibits a similar trend in the flowering stage, with CWR ranging from 92 to 139.32 mm/stage and PET from 200.6 to 208.8 mm/stage. However, the subsequent stages diverge, particularly in young fruit development, where CR-F3 displays higher CWR (33.6 mm/stage) than CR-F4 (35.2 mm/stage), suggesting varied water requirements for optimal fruit maturation. Harvest stages

in both farms demonstrate moderate CWR and PET, emphasizing the need for careful water resource allocation during these stages to ensure efficient resource utilization and crop productivity.

In the case of Sisaket coffee farms, there were different requirements for the Robusta coffee. Starting with S-F1 (Fig. 7(k)), after pruning, the substantial increase in CWR from 26 to 36.22 mm/stage reflects the heightened water demand to support the initial regrowth and new shoot development stages. This aligns with a simultaneous rise in PET from 36.4 to 39.7 mm/stage, emphasizing the need for an adequate water supply to meet the evapotranspiration demands of the recovering coffee plants. Moving to S-F2 (Fig. 7(l)), the growth stages exhibit a similar trend with a noticeable surge in CWR and PET after pruning, reaching 27.7 and 37.08 mm/stage, respectively. In S-F3 (Fig. 7(m)), the after-pruning stage registers CWR and PET values of 33.6 and 41.8 mm/stage, indicating a more pronounced demand for water during the initial growth phase. The subsequent flowering and fruit development stages maintain elevated water requirements, emphasizing the significance of precise irrigation strategies to support optimal crop development. The fluctuations in water demand throughout the growth stages signify the

dynamic nature of Arabica coffee plants' water requirements. Tailored irrigation practices become imperative to synchronize water supply with the varying needs of the crop, ensuring efficient water use. Additionally, the observed decline in water requirements during the harvest stages suggests a strategic reduction in irrigation, aligning with the natural reduction in the plant's metabolic activities as it approaches the end of its growth cycle.

The detailed analysis of the average CWR and PET data over the past five years (2018–2022) for the six Robusta coffee farms in Nan province, alongside the downscaled rainfall projections for 2023, 2028, and 2033, provides a comprehensive understanding of the intricate dynamics governing water demand and supply across various growth stages presented in Fig. 8. Each coffee farm, represented by N-F1 to N-F6, displays unique patterns in CWR across different growth stages. Notably, the flowering and fruit development stages consistently exhibit higher water requirements, underlining their significance in the coffee plant's life cycle. Firstly, focusing on CWR, it is noteworthy that each coffee farm (N-F1 to N-F6) exhibits distinct patterns in water requirements during different growth stages. For instance, in N-F1 (Fig. 8(a)), the flowering

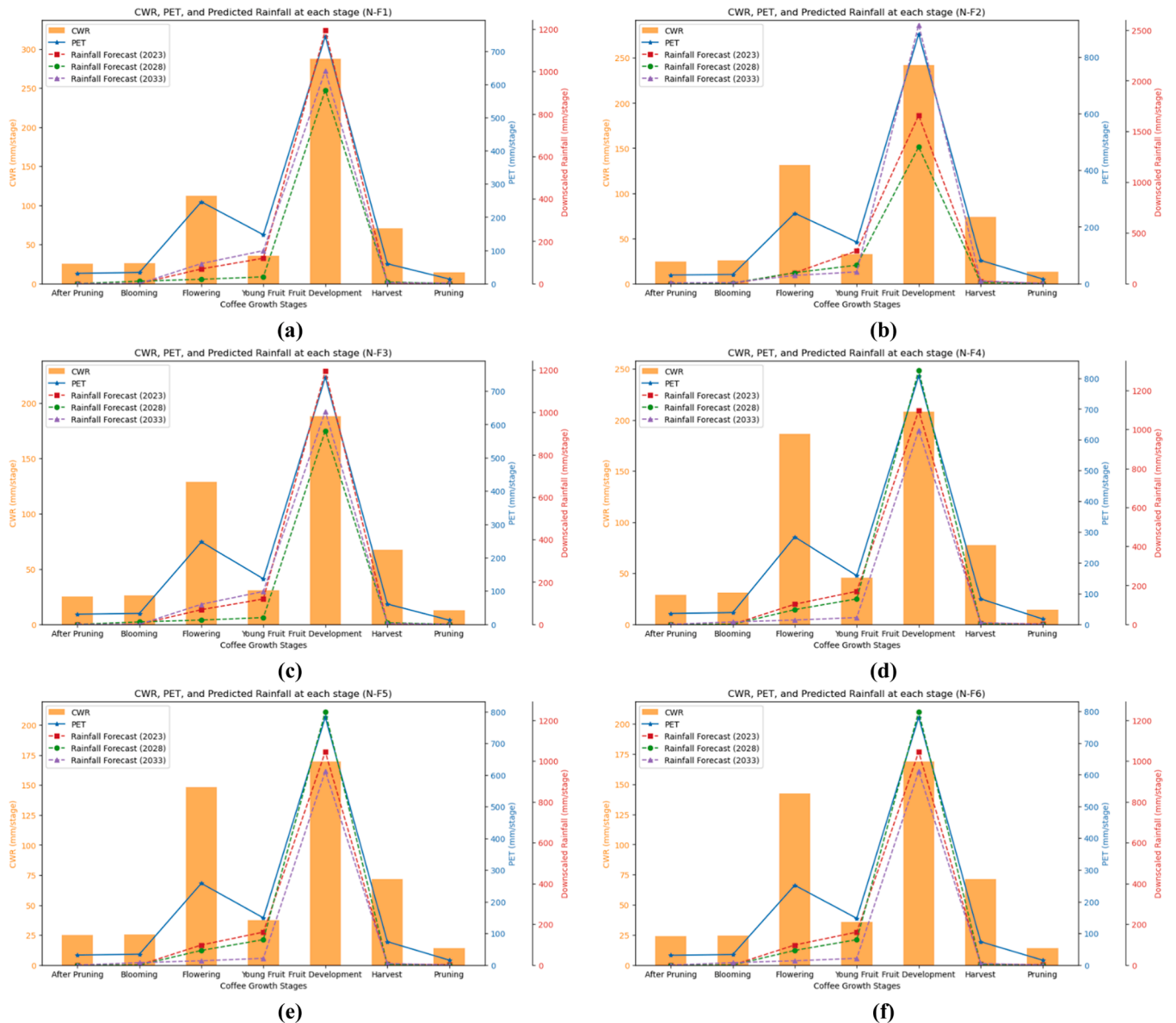


Fig. 8. Assessment of Crop Water Requirement (CWR), Potential Evapotranspiration (PET), and Downscaled Rainfall for Future Projections Across Selected Coffee Farms in Nan for the Years 2023, 2028, and 2033: (a) Coffee Farm 1, (b) Coffee Farm 2, (c) Coffee Farm 3, (d) Coffee Farm 4, (e) Coffee Farm 5, and (f) Coffee Farm 6.

stage demands a considerable CWR, ranging from 40 to 164.9 mm/stage, with an average of 112.5 mm/stage. The subsequent fruit development stage witnesses a significant surge in CWR, ranging from 120.4 to 741.7 mm/stage. It emphasizes the critical water needs during the maturation of coffee fruits. Similarly, the other farms demonstrate specific water requirement patterns, reflecting the unique characteristics of each growth stage. In parallel, examining the downscaled rainfall data for the forecasted years (2023, 2028, 2033) reveals temporal variability in precipitation patterns. These projections offer insights into potential shifts in climate conditions that may impact water availability for coffee cultivation. For example, in Fig. 8(b), examining the data for N-F2 during the fruit development stage, the downscaled rainfall is projected to be 1.18 mm/stage for 2023, indicating a relatively lower water availability during this critical phase. However, for 2033, the projection increases to 7.83 mm/stage, suggesting a potential shift in precipitation patterns. It highlights the importance of considering downscaled rainfall data for informed irrigation management decisions. Furthermore, comparing CWR with downscaled rainfall for each year elucidates the potential implications for water availability during crucial growth stages.

In N-F3, for example (Fig. 8(c)), the flowering stage shows a CWR range from 36.6 to 206.8 mm/stage, while the downscaled rainfall projections for 2023 and 2033 are 0.2 mm/stage and 0.29 mm/stage, respectively. This incongruence underscores the necessity for supplemental irrigation to meet the substantial water demand during flowering. The integration of CWR and downscaled rainfall data for each growth stage and forecasted year enhances the precision of water management strategies. For instance, in N-F4 during the fruit development stage, the CWR ranges from 152.8 to 844.2 mm/stage, while the downscaled rainfall projections for 2023 and 2033 are 0.22 mm/stage and 0.32 mm/stage, respectively (Fig. 8(d)). This detailed interpretation underscores the significance of considering CWR and downscaled rainfall data for strategic water management in coffee cultivation. The

nuanced analysis of these parameters across different farms and growth stages provides valuable insights for optimizing irrigation practices and mitigating potential water stress. The integration of downscaled rainfall projections enhances the foresight into the challenges posed by future climate scenarios, enabling proactive measures for sustainable coffee cultivation in Nan province.

Analyzing data from Arabica coffee farms in Chiang Rai reveals intricate insights into the interplay of water management, climate scenarios, and crop growth. The analysis unravels discernible variations in CWR across the four farms (CR-F1 to CR-F4). Detailed results are presented in Fig. 9. Notably, the flowering and fruit development stages consistently demand elevated CWR, signifying their pivotal role in the Arabica coffee growth cycle. For instance, CR-F1 (Fig. 9(a)) exhibits an average CWR of 166.42 mm/stage during the fruit development stage. Further analysis within each farm underscores substantial variation in CWR among distinct growth stages, with the fruit development phase consistently requiring the highest water input, registering average CWR values ranging from 651.66 mm/stage to 826.52 mm/stage across the four farms. The temporal variability in precipitation patterns is highlighted through the downscaled rainfall data for the projected years (2023, 2028, 2033). This temporal variability assumes significance in anticipating shifts in climate conditions and their potential influence on water availability for Arabica coffee cultivation. Each coffee farm exhibits distinctive responses to downscaled rainfall, accentuating the need for farm-specific adaptations to changing precipitation patterns. For example, CR-F3 during the flowering stage projects a consistent increase in rainfall from 1.28 mm/stage in 2023 to 2.29 mm/stage in 2033 (Fig. 9(c)). The juxtaposition of CWR and downscaled rainfall data reveals potential water deficits or surpluses. For instance, in CR-F2 during the fruit development stage, the CWR is 651.66 mm/stage, while the downscaled rainfall projections for 2023 and 2033 are 1 mm/stage and 4.57 mm/stage, respectively (Fig. 9(b)). These disparities underscore the necessity for supplementary irrigation strategies during

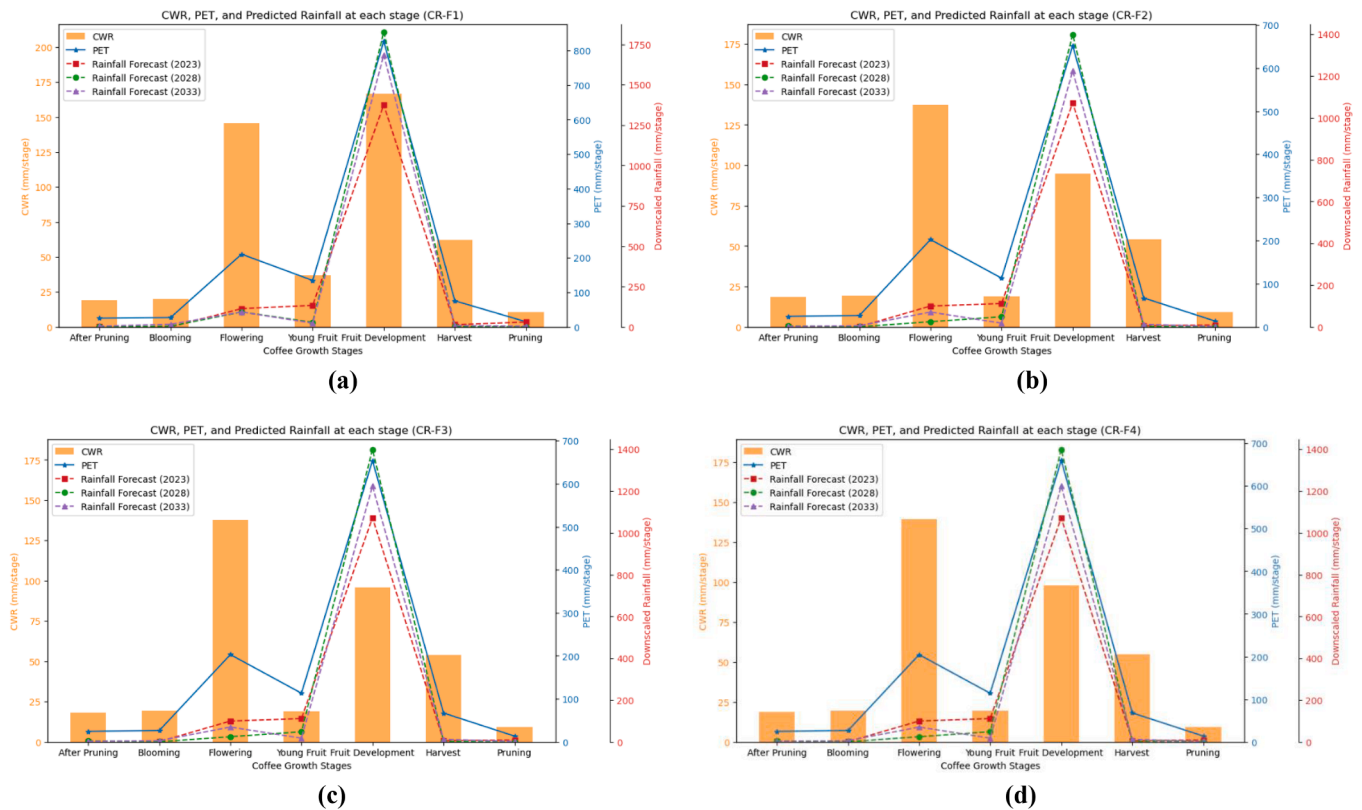


Fig. 9. Assessment of Crop Water Requirement (CWR), Potential Evapotranspiration (PET), and Downscaled Rainfall for Future Projections Across Selected Coffee Farms in Chiang Rai for the Years 2023, 2028, and 2033: (a) Coffee Farm 1, (b) Coffee Farm 2, (c) Coffee Farm 3, (d) Coffee Farm 4.

critical growth phases. Integrating CWR and downscaled rainfall data is crucial in identifying periods with heightened water demand and potential shortages, necessitating strategic irrigation planning and adopting water-efficient technologies to mitigate risks associated with unfavorable climate scenarios. The detailed examination of CWR, PET, and downscaled rainfall data lays the foundation for adaptive water management strategies. Farmers can tailor their irrigation practices to align with the specific requirements of different growth stages, minimizing water wastage and ensuring optimal crop health. The anticipation of future climate scenarios through downscaled rainfall projections enables the development of climate-resilient agricultural practices. This foresight is indispensable for sustaining Arabica coffee production in changing environmental conditions. This information is instrumental for making informed decisions in crop and water management, contributing to the long-term sustainability of coffee cultivation in Chiang Rai.

The comprehensive data analysis for Robusta coffee fields in Sisaket provides valuable insights into the CWR, PET, and downscaled rainfall patterns. Examining the growth stages reveals noteworthy trends and potential impacts of future climate scenarios on coffee cultivation. In the "After Pruning" stage, CWR ranges from 26 to 33.6 mm/stage, with evapotranspiration between 36.22 and 41.8 mm/stage. The downscaled rainfall shows variability, impacting water availability for this critical stage. In the "Blooming" stage, CWR ranges from 25.3 to 33.9 mm/stage, with evapotranspiration between 36.96 and 42.56 mm/stage. The forecasted downscaled rainfall influences the water balance, potentially affecting flowering success. Moving to the "Flowering" stage, CWR spans 76.8 to 114.2 mm/stage, and evapotranspiration ranges from 187.02 to 201.86 mm/stage. This stage is crucial for fruit setting, and the downscaled rainfall patterns predict challenges in maintaining optimal water levels. In the "Young Fruit" stage, CWR varies from 5.9 to 11.1 mm/stage, and evapotranspiration ranges from 33.86 to 41.54 mm/stage. The downscaled rainfall projections indicate potential water stress during this growth phase. The "Fruit Development" stage exhibits significant CWR variability (166.4 to 193.4 mm/stage) and evapotranspiration (234.98 to 273.72 mm/stage). The downscaled rainfall becomes a critical factor in supporting robust fruit development. The "Harvest" stage, with CWR ranging from 91.2 to 125.7 mm/stage and evapotranspiration from 129.56 to 156.2 mm/stage, highlights the importance of sufficient water availability for quality yield. In the "Pruning" stage, CWR ranges from 22.6 to 29.5 mm/stage, with evapotranspiration between 27.66 and 33.92 mm/stage. The downscaled rainfall projections play a role in shaping pruning practices. The 2023, 2028, and 2033 forecasts underscore the need for adaptive water management strategies to mitigate potential challenges posed by changing climate scenarios across all growth stages.

Settia et al. (2022) highlighted the pivotal role of rainfall as a fundamental input in diverse water management and hydrological investigations related to agriculture and crop management [51]. The quantity of water essential to offset the loss from evapotranspiration in a cultivated field is termed CWR. Although PET and CWR values may be comparable, the distinction lies in their definitions: CWR signifies the quantity of water to be provided, whereas PET denotes the amount of water lost through evapotranspiration [35]. Variations in local weather parameters and the specific crop under cultivation can contribute to discrepancies between these values. In the context of Thailand, the authors noted the absence of past investigations concerning the potential impact of CC on CWR and PET, specifically for coffee, in the study area of this research. While studies addressing this aspect exist in non-tropical environments, a notable research gap exists for the unique agricultural conditions prevalent in northern and northeastern Thailand. This research's findings align with other studies [25,32,52–55]. In the analysis of CWR for coffee Arabica (*Coffea arabica* L.), Wringley (1988) underscores the necessity of an annual range between 1200 and 1600 mm, which is crucial for ensuring optimal growth and sustainable production [32]. Fig. 9 (Chiang Rai) illustrates that CWR, PET, and future projected precipitation collectively meet a significant

portion of the water demands of coffee cultivation across all coffee farms. However, a nuanced irrigation schedule, particularly during non-rainy seasons in Thailand, becomes imperative, as future projections indicate that rainfall alone may not suffice to fulfill the evolving water requirements of coffee crops. Similar trends were observed with differences in the other coffee farms in Nan and Sisaket provinces (Figs. 8 and 10). Eriyagama, Chemin, and Alankara (2014) observed a potential reduction in effective mean annual rainfall across several significant coffee-producing regions by the 2050s [55]. CC may lead to migration and a reduction in areas conducive to coffee cultivation, as highlighted in various studies, e.g. [56,57]. The findings of this study, derived from projected precipitation data, substantiate their observation of diminishing rainfall trends within current and future CC scenarios.

4. Conclusion and future directions

In the realm of CC impact assessment on precipitation and its implications for CWR, this study conducted a rigorous evaluation of three distinct bias correction methods—DT, EQM, and QM—applied to statistical downscaling of precipitation data derived from the CanESM5 CMIP6-GCM. The findings decisively favored the DT over EQM and QM, as evidenced by its consistently superior performance across multiple key metrics. The downscaled precipitation projections from 2023 to 2055 reveal dynamic mean precipitation and SD trends across coffee farms. The analysis of farms such as N-F1, CR-F1, N-F4, S-F1, and others underscores the variability in water availability, presenting challenges related to excess and scarcity. The need for flexible irrigation schedules in response to changing precipitation, efficient water conservation practices, and the implementation of robust irrigation systems is emphasized and tailored to the specific needs of each farm. The assessment of CWR and PET during different growth stages further elucidates the intricate dynamics governing water demand in coffee farms. Variations in CWR and PET across blooming, flowering, fruit development, and harvest stages highlight the necessity for adaptive water management strategies. Detailed analyses were conducted for various coffee farms, including N-F1 to N-F6, CR-F1 to CR-F4, and S-F1 to S-F3.

For N-F1, CWR ranged from 16.7 to 33.7 mm/stage during the blooming stage, averaging 26.34 mm/stage, with a 7.92 mm/stage SD. The flowering stage demonstrated a wider range, averaging 112.5 mm/stage, with a substantial SD of 60.86 mm/stage. Fruit development exhibited the highest CWR, ranging from 120.4 to 741.7 mm/stage, averaging 287.32 mm/stage, with a significant SD of 256.41 mm/stage. These patterns were echoed in other coffee farms, illustrating the dynamic interaction between CWR and PET across various growth stages. For example, in CR-F1, the flowering stage exhibited a substantial CWR range from 100.3 to 211.9 mm/stage, averaging 145.7 mm/stage with a notable SD of 43.57 mm/stage. Fruit development stages demonstrated a wide-ranging CWR, varying from 51.4 to 900.3 mm/stage. Similar detailed assessments were conducted for other farms, elucidating nuanced water requirements during growth stages. In CR-F2, the flowering stage stood out with a substantial CWR range of 89.7 to 198.6 mm/stage, averaging 137.32 mm/stage and exhibiting a relatively low SD of 40 mm/stage. Contrastingly, the blooming stage showcased a diverse CWR range from 0.6 to 27 mm/stage, emphasizing the dynamic water needs during this phase. The analysis extended to Sisaket coffee farms, with distinct requirements for Robusta coffee. Noteworthy trends were observed, such as a substantial increase in CWR after pruning, emphasizing the heightened water demand to support initial regrowth and new shoot development stages. Similar trends were identified in various growth stages across different farms, underscoring the dynamic nature of water requirements for Arabica and Robusta coffee plants. Additionally, integrating downscaled rainfall projections for 2023, 2028, and 2033 provided a comprehensive understanding of potential shifts in precipitation patterns. The comparison of CWR with downscaled rainfall data revealed incongruences, emphasizing the need for supplementary

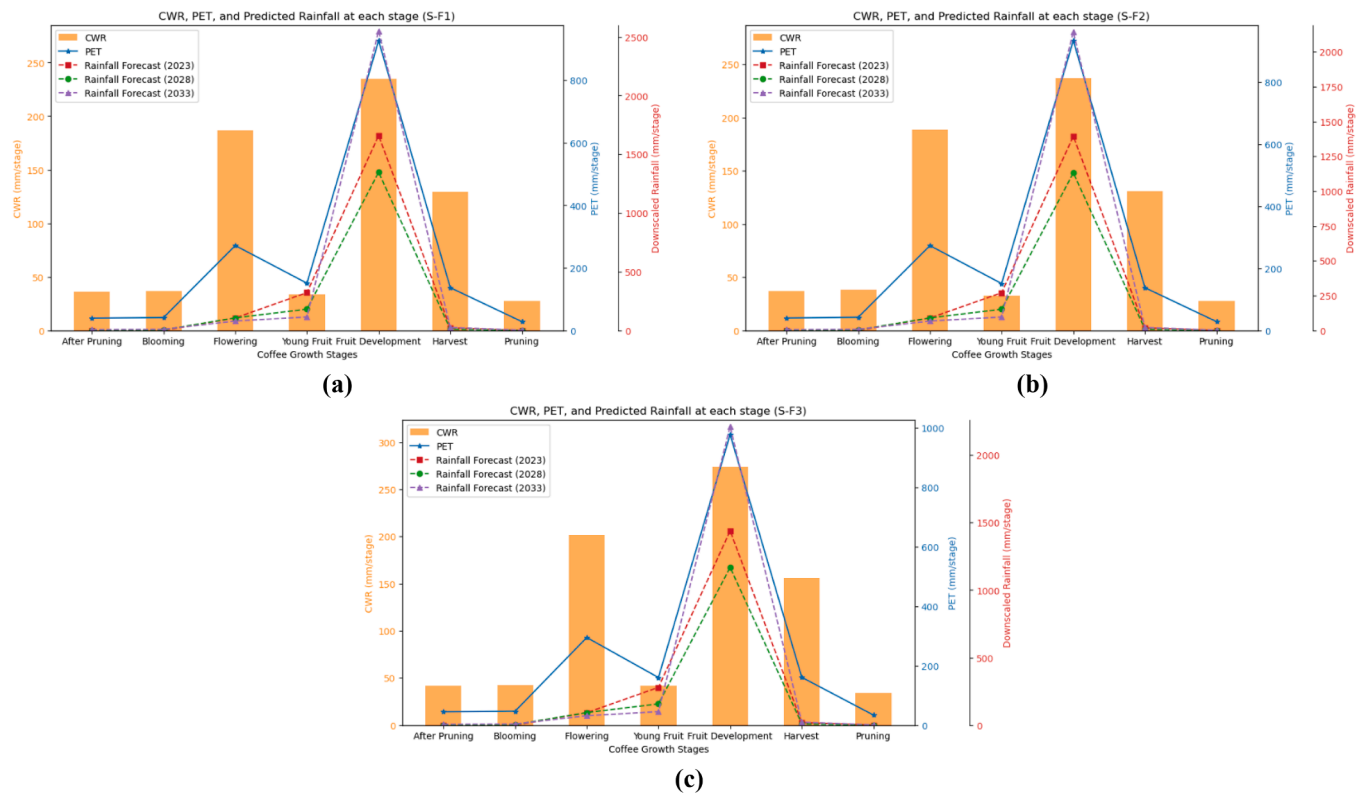


Fig. 10. Assessment of Crop Water Requirement (CWR), Potential Evapotranspiration (PET), and Downscaled Rainfall for Future Projections Across Selected Coffee Farms in Sisaket for the Years 2023, 2028, and 2033: (a) Coffee Farm 1, (b) Coffee Farm 2, (c) Coffee Farm 3.

irrigation strategies during critical growth phases. The anticipated impacts of CC on Arabica and Robusta coffee farms in Chiang Rai and Sisaket, respectively, present additional challenges and opportunities for water management. Examining CWR, PET, and downscaled rainfall data for Arabica coffee farms in Chiang Rai emphasizes the importance of integrating climatic projections into adaptive water management practices. The distinct responses of CR-F1 to CR-F4 to changing precipitation patterns underscore the need for farm-specific strategies to address potential water deficits or surpluses during critical growth phases. Similarly, the analysis of Robusta coffee fields in Sisaket reveals the dynamic nature of water requirements across various growth stages, necessitating adaptive strategies to mitigate the impacts of changing climate scenarios. This research highlights the importance of tailored, farm-specific approaches in mitigating the impacts of climate change. Future research could delve into refining bias correction methods, enhancing spatial and temporal resolutions, incorporating multiple climate models, and integrating dynamic crop models for a more comprehensive understanding of the intricate dynamics governing coffee cultivation under the influence of CC.

Funding

This research received no external funding.

Ethics statement

Not applicable: This manuscript does not include human or animal research.

CRedit authorship contribution statement

Usa Wannasingha Humphries: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Muhammad Waqas:** Writing – original draft, Methodology, Investigation, Data

curation, Conceptualization. **Phyo Thandar Hlaing:** Visualization, Validation, Software. **Angkool Wangwongchai:** Writing – review & editing, Supervision, Methodology, Investigation. **Porntip Dechpichai:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research project is supported by King Mongkut’s University of Technology Thonburi (KMUTT), Thailand Science Research and Innovation (TSRI), and the National Science, Research and Innovation Fund (NSRF) Fiscal year 2024.

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